

Nitrite-N in Seawater

1 SCOPE

This method covers the determination of Nitrite-N in Seawater.

2 RANGE OF APPLICATION

Range 0.07 μM – 14 μM (1.0 – 200 $\mu\text{g N/L}$)

3 METHOD DETECTION LIMIT

By EPA Procedure MDL = 0.023 μM (0.31 $\mu\text{g N/L}$)

4 METHOD PRINCIPLE

Nitrite (NO_2^-) is determined through formation of a reddish purple azo dye produced at pH 2.0 to 2.5 by coupling diazotized sulfanilamide with *N*-(1-naphthyl)-ethylenediamine dihydrochloride (NEDD). The absorbance of this complex is measured spectrophotometrically at 520 nm.

5 REFERENCES

Methods for the Determination of Inorganic Substances in Environmental Samples, EPA 600/R 93/100, August 1993: Method 353.2, Revision 2.0.

Standard Methods for the Examination of Water and Wastewater, APHA/AWWA/WEF, method 4500 NO_2^- B (2000).

6 INTERFERENCES

There are few significant interferences at concentrations less than 1000 times that of the nitrite. Presence of strong oxidants or reductants in the samples may affect the nitrite determination.

Highly alkaline samples may give low results due to the shift in pH.

Bias from sample turbidity or color is correctable, using the sample blanking feature of the AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the safety data sheets (SDS) for details on safe handling of chemicals.

<u>Chemical</u>	<u>Formula</u>	<u>C.A.S. No.</u>	<u>Safety Classification</u>
N-(1-naphthyl)-ethylenediamine dihydrochloride	$C_{12}H_{14}N_2 \cdot 2HCl$	1465-25-4	GHS07
Hydrochloric acid, concentrated*	HCl	7647-01-0	GHS05, GHS07
Phosphoric acid, concentrated (low in nitrite)	H_3PO_4	7664-38-2	GHS05
Sodium hydroxide	NaOH	1310-73-2	GHS05
Sodium nitrite	$NaNO_2$	7632-00-0	GHS03, GHS06, GHS09
Sodium chloride ⁺	NaCl	7647-14-5	GHS05
Sulfanilamide	$C_6H_8N_2O_2S$	63-74-1	--

GHS01: Danger – Explosive

GHS02: Danger – Flammable

GHS03: Danger – Oxidizing

GHS04: Warning – Compressed Gas

GHS05: Warning/Danger – Corrosive

GHS06: Danger – Toxic

GHS07: Warning – Irritant

GHS08: Danger – Health Hazard

GHS09: Warning/Danger – Aquatic Toxicity

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

* Only required for alternative color reagent.

⁺ Only required for synthetic seawater preparation.

8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 SULFANILAMIDE-NEDD

Sodium hydroxide	0.5 g
Phosphoric acid, concentrated	10 mL
Sulfanilamide	3.75 g
N-(1-naphthyl)-ethylenediamine dihydrochloride	0.19 g
DI Water	dilute to 500 mL

Dissolve 0.5 g sodium hydroxide pellets in about 250 mL deionized water. Carefully add 10 mL phosphoric acid, concentrated (low in nitrite). Add 3.75 g sulfanilamide and 0.19 g N-(1-naphthyl)-ethylenediamine dihydrochloride and stir to dissolve. Dilute to 500 mL with deionized water. Filter this reagent, e.g., using filter paper, such as Whatman #1 filter paper. As the reagent slowly turns pink, re-filter as needed.

Solution is stable for 1 month when refrigerated in a dark bottle. Discard when background absorbance of analyzed blanks exceeds 0.01 AU.

*** ALTERNATIVE RECIPE***

If there is concern about the phosphoric acid in the color reagent, then an alternative recipe using hydrochloric acid can be used.

Hydrochloric acid, concentrated	15 mL
Sulfanilamide	2.0 g
Sodium hydroxide, pellets	1.9 g
N-(1-naphthyl)-ethylenediamine dihydrochloride	0.100 g
DI Water	dilute to 500 mL

Carefully add 15 mL concentrated phosphoric acid (low in nitrite) to approximately 300 mL deionized water. Add 2.0 g sulfanilamide, 1.9 g of sodium hydroxide (pellets) and stir until dissolved. Add 0.100 g N-(1-naphthyl)-ethylenediamine dihydrochloride and stir until dissolved. Dilute to 500 mL with deionized water and store in a dark bottle.

This reagent is stable for at least 1 month when stored at 4°C. Discard if reagent possesses or develops pink color. For best performance, filter this reagent at time of use.

9 STANDARD SOLUTIONS

9.1 NITRITE-N STOCK STANDARD SOLUTION (35.7 mM / 500 mg N/L)

Sodium nitrite	2.463 g
DI water	dilute to 1 L

Dissolve 2.463 g sodium nitrite (dried at 105°C) in about 800 mL deionized water in a 1 liter volumetric flask. Dilute to the mark with deionized water.

Store up to 1 month in an amber bottle at 4°C.

9.2 CALIBRATION DILUENT (SYNTHETIC SEAWATER)

Sodium chloride	35 g
DI water	dilute to 1 L

Dissolve 35 g of sodium chloride in about 800 mL of deionized water in a 1 liter volumetric flask. Dilute to 1 liter with deionized water.

***Note:** The concentration of sodium chloride can be adjusted to match the salinity of the seawater samples. Low nutrient seawater is also available.

Store at room temperature. Stable for 1 month.

9.3 NITRITE-N INTERMEDIATE STANDARD SOLUTION (142.86 µM / 2000 µg N/L)

NITRITE-N Stock Standard (35.7 mM N/L, above)	1.0 mL
Calibration diluent	dilute to 250 mL

To a 250 mL volumetric flask pipet 1.0 mL NITRITE-N stock standard (35.7 mM N/L) and dilute to the mark with calibration diluent.

Prepare weekly.

9.4 AUTO CALIBRATION STANDARD – NITRITE-N (14.29 µM / 200 µg N/L)

NITRITE-N Intermediate Standard (142.86 µM N/L, above)	10 mL
Calibration diluent	dilute to 100 mL

To a 100 mL volumetric flask pipet 10 mL NITRITE-N intermediate standard (142.86 µM N/L) and dilute to the mark with calibration diluent.

Prepare fresh daily.

9.5 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, µM	0	0.07	0.14	0.36	0.71	1.79	3.57	5.36	10.72	14.29
Volume (mL) Auto Calibration Standard Solution diluted as indicated with Calibration diluent	--	0.5	1.0	2.5	5.0	12.5	25	37.5	75	100
Final Volume, mL	100	100	100	100	100	100	100	100	100	100

Prepare daily.

10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected into plastic or glass containers.

The laboratory should consult the most currently acceptable sample storage and preservation procedures as mandated by the EPA.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Any reagents having visible particulates should be filtered.

Pour calibration standards, quality control solutions and samples into sample cups and reagent wedges, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent wedges and a waste collection bottle in place.

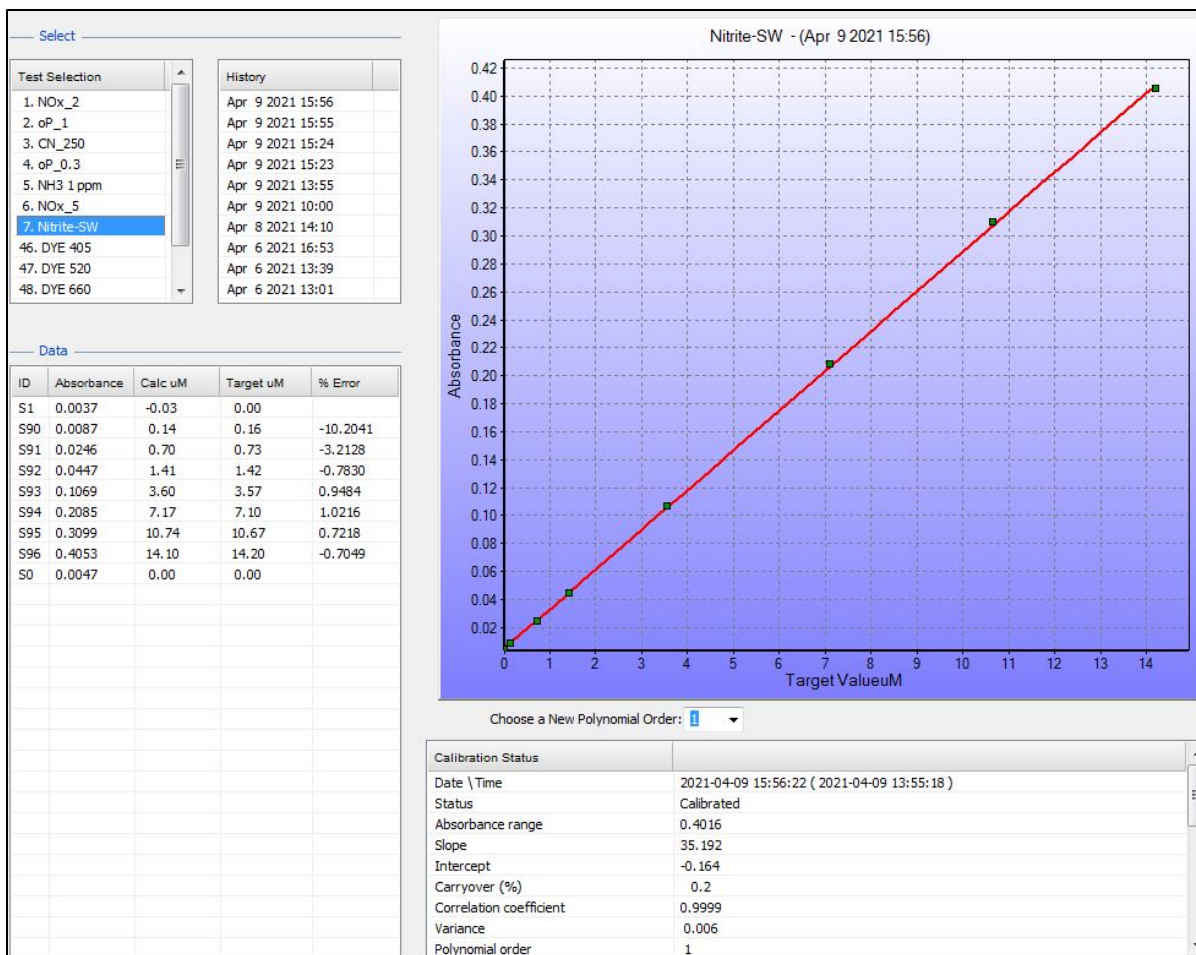
12 TEST PARAMETERS

PARAMETER	AQ SETTING
Test name	Nitrite 200
Units	μM
Decimals	2
Test type	End point
Sample volume (μL)	450
Water volume (μL)	50
Number of mixes	2
Cuvette primes	1
Cuvette washes	2
Baseline on Wash	Ticked
Reaction time (seconds)	360
Wavelength (nm)	520
Polynomial order	1
Number of reagents	1
1. Sulfanilamide – NEDD (μL)	50
Advanced Test Parameters	☒ Eliminate Air from Test Transfer ☒ Extra Debubbling Action

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.01 AU

Sensitivity Range (14.2 μ M N): 0.32 – 0.48 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover (%): Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A - B)}{C} \times 100\%$$

Where, A = absorbance of carryover blank (S0)
 B = absorbance of blank (S1)
 C = calibration absorbance range

14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.07 μM (1.0 $\mu\text{g N/L}$) nitrite standard in synthetic seawater***

	Concentration (μM)			
	0.06	0.05	0.08	0.06
	0.07	0.05	0.05	0.05
	0.06	0.06	0.05	0.06
	0.05	0.07	0.06	0.07
	0.05	0.05	0.04	0.07
	0.06	0.06	0.04	0.07
	0.06	0.06	0.05	0.07
	0.06	0.07	0.06	0.07
	0.05	0.07	0.05	0.06
	0.06	0.04	0.06	0.06
	0.07	0.06	0.06	0.06
	0.06	0.06	0.06	0.04
	0.05	0.06	0.07	0.05
	0.06	0.06	0.07	0.06
	0.07	0.07	0.08	0.07
Statistics				
Mean	0.06	0.06	0.06	0.06
Std. Deviation	0.01	0.01	0.01	0.01
MDL (μM)*	0.02	0.02	0.03	0.02
MDL ($\mu\text{g N/L}$)	0.24	0.31	0.42	0.30

14.2 PERFORMANCE STUDY

Replicates of 0.71 μM (10.0 $\mu\text{g N/L}$) nitrite standard in synthetic seawater***

	Concentration (μM)			
	0.70	0.65	0.68	0.70
	0.69	0.70	0.69	0.71
	0.71	0.66	0.70	0.71
	0.71	0.70	0.71	0.67
	0.72	0.69	0.70	0.68
	0.70	0.66	0.71	0.70
	0.69	0.69	0.69	0.72
	0.70	0.67	0.70	0.72
	0.72	0.66	0.70	0.69
	0.72	0.68	0.72	0.68
Statistics				
Mean	0.71	0.68	0.70	0.70
Std. Deviation	0.01	0.02	0.01	0.02
% RSD**	1.6	2.6	1.8	2.5
% Recovery	99.5	95.3	98.5	98.4

14.3 PRECISION STUDY

Replicates of 7.14 μM (100.0 $\mu\text{g N/L}$) nitrite standard in synthetic seawater***

Concentration (μM)				
Statistics				
Mean	7.08	7.12	7.25	7.05
% Recovery	99.1	99.7	101.5	98.7
Std. Deviation	0.06	0.04	0.05	0.04
% RSD**	0.8	0.5	0.6	0.6
Maximum	7.21	7.17	7.31	7.12
Minimum	7.00	7.06	7.18	6.96

Replicates of 14.29 μM (200.0 $\mu\text{g N/L}$) nitrite standard in synthetic seawater**

Concentration (μM)				
Statistics				
Mean	14.25	14.23	14.35	14.22
% Recovery	99.7	99.6	100.4	99.5
Std. Deviation	0.11	0.07	0.10	0.08
% RSD**	0.8	0.5	0.7	0.5
Maximum	14.40	14.37	14.54	14.36
Minimum	14.03	14.13	14.23	14.10

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9. Synthetic seawater contains 35 g/L of NaCl.

15 REVISIONS

April 2021

Rev 0

Method created and support data acquired